Paracetamol and its side effects

Paracetamol,[a] or **acetaminophen**,[b] is an analgesic and antipyretic agent used to treat fever and mild to moderate pain.[14][15] It is a widely available over-the-counter drug sold generically or under various brand names, including **Tylenol** and **Panadol**.

Paracetamol relieves pain in both acute mild migraine and episodic tension headache.[16][17] At a standard dose, paracetamol slightly reduces fever,[14][18][19] though it is inferior to ibuprofen in that respect[20] and the benefits of its use for fever are unclear, particularly in the context of fever of viral origins.[21] The aspirin/paracetamol/caffeine combination also helps with both conditions when the pain is mild and is recommended as a first-line treatment for them.[22] Paracetamol is effective for pain after wisdom tooth extraction, but it is less effective than ibuprofen.[23] The combination of paracetamol and ibuprofen provides greater analgesic efficacy than either drug alone.[23][24] The pain relief paracetamol provides in osteoarthritis is small and clinically insignificant.[15][25] Evidence supporting its use in low back pain, cancer pain, and neuropathic pain is insufficient.[15][26][27][28][29][30] Paracetamol is the first-line treatment for pain and fever in pregnancy; no causal association with neurodevelopmental disorders has been established, while untreated pain and fever can harm the mother and fetus.[31][32][33]

In support of existing recommendations for its safe use during pregnancy, a 2026 review found that there were no clinically significant increases in the occurrence of autism, attention deficit hyperactivity disorder, or intellectual disability in the children of pregnant women who used paracetamol as directed.[34]

In the short term, paracetamol is safe and effective when used as directed.[35] Short term adverse effects are uncommon and similar to ibuprofen.[36] Paracetamol is typically safer than nonsteroidal anti-inflammatory drugs (NSAIDs) for long-term use.[37] Paracetamol is also often used in patients who cannot tolerate NSAIDs like ibuprofen.[38][39] Chronic consumption of paracetamol may result in a drop in hemoglobin level, indicating possible gastrointestinal bleeding,[40] and abnormal liver function tests. The recommended maximum daily dose for an adult is three to four grams.[26][41] Higher doses may lead to toxicity, including liver failure.[42] Paracetamol poisoning is the foremost cause of acute liver failure in the Western world, and accounts for most drug overdoses in the United States, the United Kingdom, Australia, and New Zealand.[43][44][45]

Paracetamol was first made in 1878 by Harmon Northrop Morse or possibly in 1852 by Charles Frédéric Gerhardt.[46][47][48] It is the most commonly used medication for pain and fever in both the United States and Europe.[49] It is on the World Health Organization's List of Essential Medicines.[50] Paracetamol is available as a generic medication, with brand names

including Tylenol and Panadol, among others.[51] In 2023, it was the 112th most commonly prescribed medication in the United States, with more than 5 million prescriptions.[52][53]

Medical uses

Fever

Paracetamol is used for reducing fever.[54] However, there has been a lack of research on its antipyretic properties, particularly in adults, and thus its benefits are unclear.[14] As a result, it has been described as over-prescribed for this application.[14] In addition, low-quality clinical data indicates that when used for the common cold, paracetamol may relieve a stuffed or runny nose, but not other cold symptoms such as sore throat, malaise, sneezing, or cough.[55]

For people in critical care, paracetamol decreases body temperature by only 0.2–0.3 °C more than control interventions and has no effect on their mortality.[18] It did not change the outcome in febrile patients with stroke.[56] The results are contradictory for paracetamol use in sepsis: higher mortality, lower mortality, and no change in mortality were all reported.[18] Paracetamol offered no benefit in the treatment of dengue fever and was accompanied by a higher rate of liver enzyme elevation: a sign of potential liver damage.[57] Overall, there is no support for a routine administration of antipyretic drugs, including paracetamol, to hospitalized patients with fever and infection.[21]

The efficacy of paracetamol in children with fever is unclear.[58] Paracetamol should not be used solely to reduce body temperature; however, it may be considered for children with fever who appear distressed.[59] It does not prevent febrile seizures.[59][60] It appears that 0.2 °C decrease of the body temperature in children after a standard dose of paracetamol is of questionable value, particularly in emergencies.[14] Based on this, some physicians advocate using higher doses that may decrease the temperature by as much as 0.7 °C.[19] Meta-analyses showed that paracetamol is less effective than ibuprofen in children (marginally less effective, according to another analysis[61]), including children younger than 2 years old,[62] with equivalent safety.[20] Exacerbation of asthma occurs with similar frequency for both medications.[63]

Pain

Paracetamol is used for the relief of mild to moderate pain such as headache, muscle aches, minor arthritis pain, and toothache, as well as pain caused by cold, flu, sprains, and dysmenorrhea.[64] It is recommended, in particular, for acute mild to moderate pain, since the evidence for the treatment of chronic pain is insufficient.[15]

Musculoskeletal pain

The benefits of paracetamol in musculoskeletal conditions, such as osteoarthritis and backache, are uncertain.[15]

It appears to provide only small and not clinically important benefits in osteoarthritis.[15][26] American College of Rheumatology and Arthritis Foundation guideline for the management of osteoarthritis notes that the effect size in clinical trials of paracetamol has been very small, which suggests that for most individuals it is ineffective.[25] The guideline conditionally recommends paracetamol for short-term and episodic use to those who do not tolerate nonsteroidal anti-inflammatory drugs. For people taking it regularly, monitoring for liver toxicity is required.[25] Essentially the same recommendation was issued by European League Against Rheumatism (EULAR) for hand osteoarthritis.[65] Similarly, the ESCEO algorithm for the treatment of knee osteoarthritis recommends limiting the use of paracetamol to short-term rescue analgesia only.[66]

Paracetamol is ineffective for acute low back pain.[15][27] No randomized clinical trials evaluated its use for chronic or radicular back pain, and the evidence in favor of paracetamol is lacking.[28][26][27]

Headaches

Paracetamol is effective for acute migraine:[16] 39% of people experience pain relief at one hour compared with 20% in the control group.[67] The aspirin/paracetamol/caffeine combination also "has strong evidence of effectiveness and can be used as a first-line treatment for migraine".[22] Paracetamol on its own only slightly alleviates episodic tension headache in those who have them frequently.[17] However, the aspirin/paracetamol/caffeine combination is superior to both paracetamol alone and placebo and offers meaningful relief of tension headache: two hours after administering the medication, 29% of those who took the combination were pain-free as compared with 21% on paracetamol and 18% on placebo.[68] The German, Austrian, and Swiss headache societies and the German Society of Neurology recommend this combination as a "highlighted" one for self-medication of tension headache, with paracetamol/caffeine combination being a "remedy of first choice", and paracetamol a "remedy of second choice".[69]

Dental and other post-surgical pain

Pain after a dental surgery provides a reliable model for the action of analgesics on other kinds of acute pain.[70] For the relief of such pain, paracetamol is inferior to ibuprofen.[23] Full therapeutic doses of nonsteroidal anti-inflammatory drugs (NSAIDs) ibuprofen, naproxen, or diclofenac are clearly more efficacious than the paracetamol/codeine combination which is frequently prescribed for dental pain.[71] The combinations of paracetamol and NSAIDs ibuprofen or diclofenac are promising, possibly offering better pain control than either

paracetamol or the NSAID alone.[23][24][72][73] Additionally, the paracetamol/ibuprofen combination may be superior to paracetamol/codeine and ibuprofen/codeine combinations.[24]

A meta-analysis of general post-surgical pain, which included dental and other surgery, showed the paracetamol/codeine combination to be more effective than paracetamol alone: it provided significant pain relief to as much as 53% of the participants, while the placebo helped only 7%.[74]

Other pain

Paracetamol fails to relieve procedural pain in newborn babies.[75][76] For perineal pain postpartum paracetamol appears to be less effective than nonsteroidal anti-inflammatory drugs (NSAIDs).[77]

The studies to support or refute the use of paracetamol for cancer pain and neuropathic pain are lacking.[29][30] There is limited evidence in favor of the use of the intravenous form of paracetamol for acute pain control in the emergency department.[78] The combination of paracetamol with caffeine is superior to paracetamol alone for the treatment of acute pain.[79]

Adverse effects

Gastrointestinal adverse effects such as nausea and abdominal pain are extremely uncommon, and their frequency is substantially lower than with ibuprofen use.[39] Increase in risk-taking behavior is possible.[93] According to the U.S. Food and Drug Administration (FDA), the drug may cause rare and possibly fatal skin reactions such as Stevens–Johnson syndrome and toxic epidermal necrolysis,[94] Rechallenge tests and an analysis of American but not French pharmacovigilance databases indicated a risk of these reactions.[94][95]

In clinical trials for osteoarthritis, the number of participants reporting adverse effects was similar for those on paracetamol and on placebo. However, the abnormal liver function tests (meaning there was some inflammation or damage to the liver) were almost four times more likely in those on paracetamol, although the clinical importance of this effect is uncertain.[96] After 13 weeks of paracetamol therapy for knee pain, a drop in hemoglobin level indicating gastrointestinal bleeding was observed in 20% of participants, this rate being similar to the ibuprofen group.[40]

Due to the absence of controlled studies, most of the information about the long-term safety of paracetamol comes from observational studies.[39] These indicate a consistent pattern of increased mortality as well as cardiovascular (stroke, myocardial infarction), gastrointestinal (ulcers, bleeding) and renal adverse effects with increased dose of paracetamol.[40][39][97] Use of

paracetamol is associated with 1.9 times higher risk of peptic ulcer.[39] Those who take it regularly at a higher dose (more than 2–3 g daily) are at much higher risk (3.6–3.7 times) of gastrointestinal bleeding and other bleeding events.[98] Meta-analyses suggest that paracetamol may increase the risk of kidney impairment by 23%[99] and kidney cancer by 28%.[97] Paracetamol slightly but significantly increases blood pressure and heart rate.[39] A review of available research has suggested that increase in systolic blood pressure and increased risk of gastrointestinal bleeding associated with chronic paracetamol use shows a degree of dose dependence.[98]

The association between paracetamol use and asthma in children has been a matter of controversy.[100] However, the most recent research suggests that there is no association,[101] and that the frequency of asthma exacerbations in children after paracetamol is the same as after another frequently used pain killer, ibuprofen.[63]

In recommended doses, the side effects of paracetamol are mild to non-existent.[102] In contrast to aspirin, it is not a blood thinner (and thus may be used in patients where bleeding is a concern), and it does not cause gastric irritation.[103] Compared to Ibuprofen—which can have adverse effects that include diarrhea, vomiting, and abdominal pain—paracetamol is well tolerated with fewer side effects.[104] Prolonged daily use may cause kidney or liver damage.[103][105] Paracetamol is metabolized by the liver and is hepatotoxic; side effects may be more likely in chronic alcoholics or patients with liver damage.[102][106]

Overdose

Main article: Paracetamol poisoning

Overdose of paracetamol is caused by taking more than the recommended maximum daily dose of paracetamol for healthy adults (three or four grams),[41] and can cause potentially fatal liver damage.[107][108] A single dose should not exceed 1000 mg, doses should be taken no sooner than four hours apart, and no more than four doses (4000 mg) in 24 hours.[41] While a majority of adult overdoses are linked to suicide attempts, many cases are accidental, often due to the use of more than one paracetamol-containing product over an extended period.[109]

Paracetamol is hepatotoxic and depletes the stock of the antioxidant glutathione in the liver as glutathione is consumed much faster than it can be replenished.

Paracetamol toxicity has become the foremost cause of acute liver failure in the United States by 2003,[45] and as of 2005, paracetamol accounted for most drug overdoses in the United States, the United Kingdom, Australia, and New Zealand.[110] As of 2004, paracetamol overdose resulted in more calls to poison control centers in the U.S. than overdose of any other pharmacological substance.[111] According to the FDA, in the United States, "56,000 emergency room visits, 26,000 hospitalizations, and 458 deaths per year [were] related to acetaminophen-

associated overdoses during the 1990s. Within these estimates, unintentional acetaminophen overdose accounted for nearly 25% of the emergency department visits, 10% of the hospitalizations, and 25% of the deaths."[112][*needs update*]

Overdoses are frequently related to high-dose recreational use of prescription opioids, as these opioids are most often combined with paracetamol.[113] The overdose risk may be heightened by frequent consumption of alcohol.[114]

Untreated paracetamol overdose results in a lengthy, painful illness. Signs and symptoms of paracetamol toxicity may initially be absent or non-specific symptoms. The first symptoms of overdose usually begin several hours after ingestion, with nausea, vomiting, sweating, and pain as acute liver failure starts.[115] People who take overdoses of paracetamol do not fall asleep or lose consciousness, although most people who attempt suicide with paracetamol wrongly believe that they will be rendered unconscious by the drug.[116][117]

Treatment is aimed at removing the paracetamol from the body and replenishing glutathione.[117] Activated charcoal can be used to decrease absorption of paracetamol if the person comes to the hospital soon after the overdose. While the antidote, acetylcysteine (also called *N*-acetylcysteine or NAC), acts as a precursor for glutathione, helping the body regenerate enough to prevent or at least decrease the possible damage to the liver; a liver transplant is often required if damage to the liver becomes severe.[43][118]

NAC is usually given following a treatment nomogram for patients with an acute overdose at a known time of ingestion.[119] Patients with unknown time of ingestion, unreliable history, or repeat supratherapeutic ingestion are treated based on risk assessment.[119] Toxicity of paracetamol is due to its quinone metabolite NAPQI and NAC also helps in neutralizing it.[117] Kidney failure is also a possible side effect.[114]